Multi Discipline Physiological Load-Recovery Model Final Report

Introduction

This study investigates the relationship between training load, physiological recovery, and performance in an endurance athlete, focusing on identifying the key predictors of Heart Rate Variability (HRV). HRV is widely used as a non-invasive indicator of autonomic nervous system activity and serves as a strong signal for recovery status, fatigue, and overall physiological readiness.

This project had two main objectives: first, to model the relationship between biological and training-derived variables and HRV over a period of time; and second, to translate these insights into a practical decision-making framework capable of informing day-to-day training adjustments.

The dataset consisted of longitudinal, athlete-specific data collected between October 25 and April 12, spanning approximately 115 days. Data sources included wearable-derived physiological metrics from a Whoop 5.0 device, as well as engineered training variables derived from cycling, running, and swimming sessions. In total, over 40 candidate features were considered, encompassing measures of cardiovascular response, training load, sleep, and subjective exertion.

Through a combination of feature engineering, linear regression modeling, and decision system design, this project aimed to bridge the gap between raw physiological data and actionable training insights in an applied endurance training context.

Data and Feature Engineering

The dataset integrated both directly measured physiological variables and engineered performance metrics to capture the multidimensional nature of training stress and recovery.

Core physiological variables—including HRV, resting heart rate (RHR), sleep duration, sleep debt, strain, and recovery score—were sourced directly from the Whoop platform. These variables provided a continuous record of autonomic recovery and systemic stress.

To quantify training load, a standardized metric—Training Stress Score (TSS)—was computed across all modalities. For cycling sessions, TSS was derived using normalized power, session duration, and intensity factor (IF), where IF was calculated as the ratio of normalized power to functional threshold power (FTP). Normalized power was obtained from power meter data and reflects a smoothed representation of effort over time.

For running and swimming, analogous calculations were performed using pace-based metrics. When sessions involved variable pacing (e.g., intervals), representative "normalized pace" values were computed programmatically, enabling estimation of intensity factor and subsequent TSS. Daily TSS was then aggregated across all activities to produce a total training load metric.

To capture the dynamics of fatigue and adaptation, rolling load metrics were constructed:

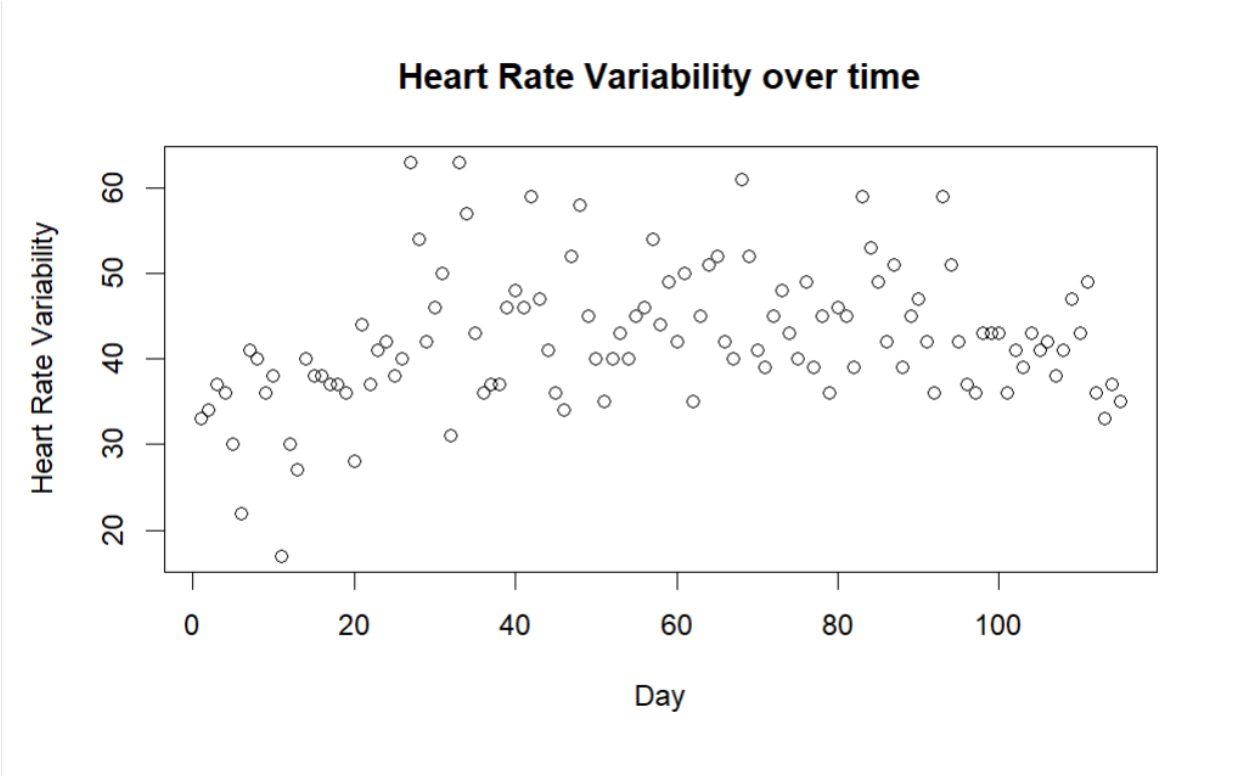
- 7-day rolling TSS (acute load)
- 28-day rolling TSS (chronic load)

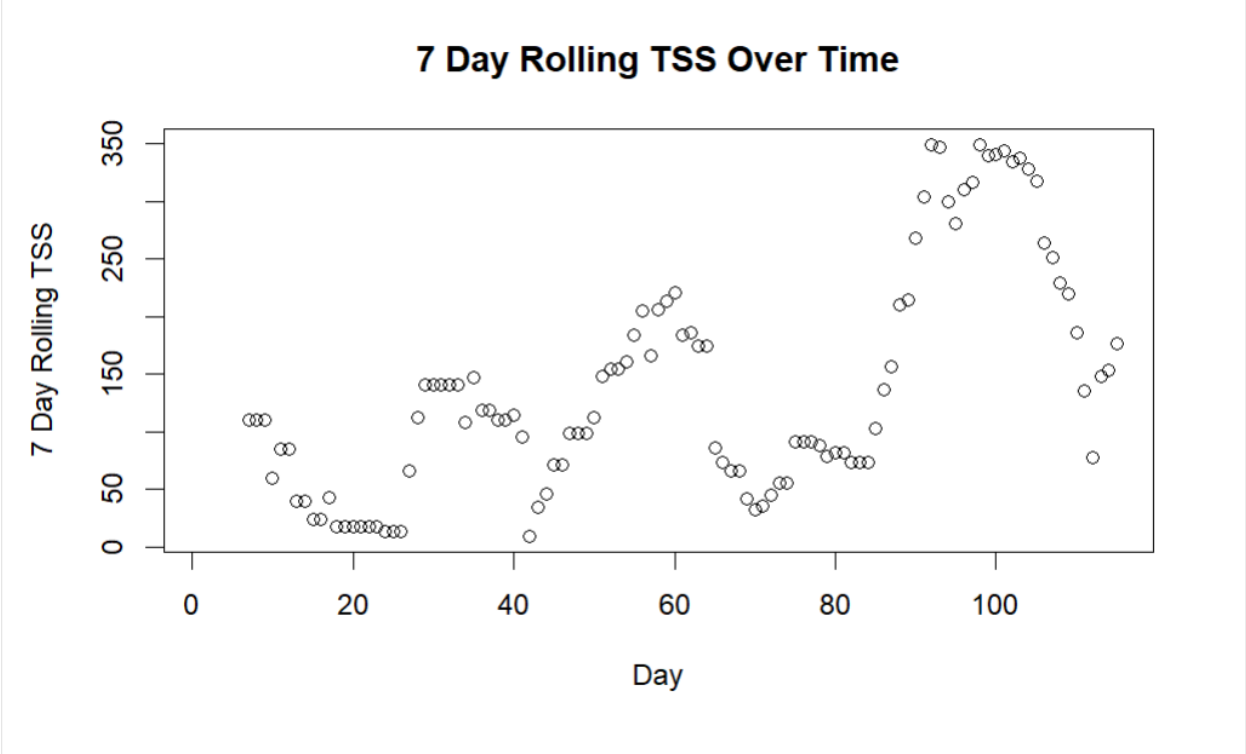
Additionally, lagged TSS variables (1–3 days) were engineered to represent delayed physiological responses to prior training stress.

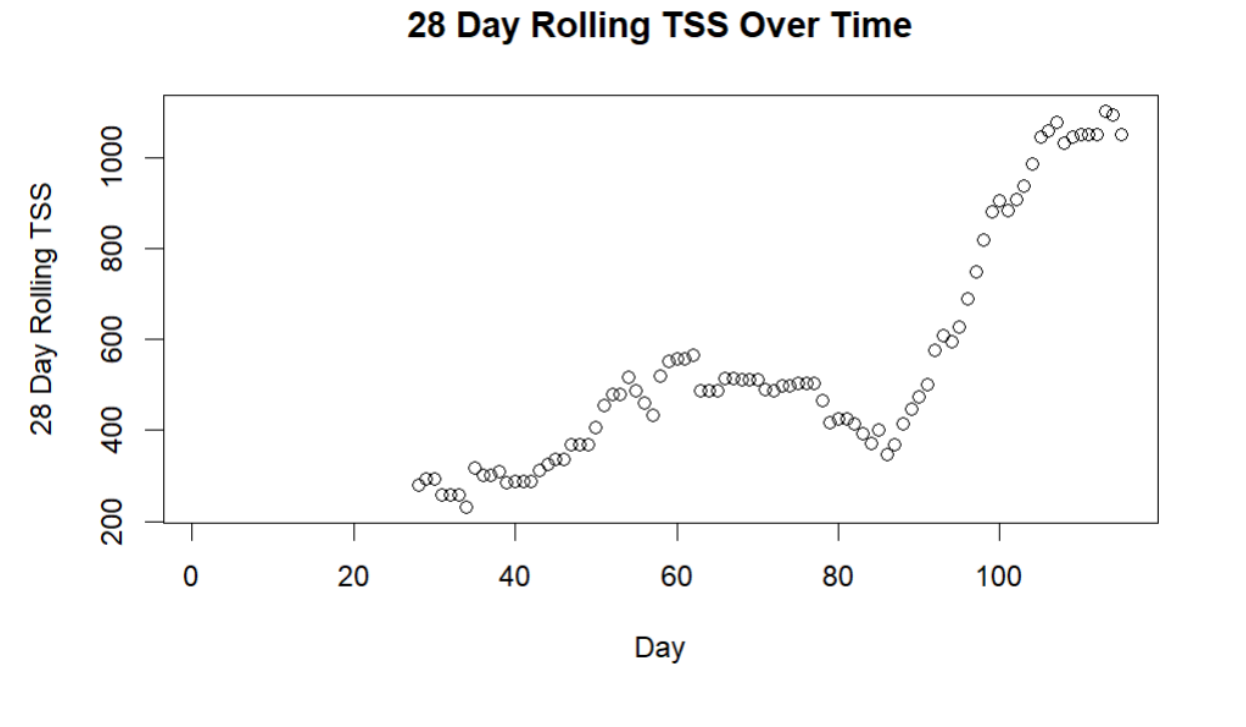
Several efficiency and adaptation metrics were also developed, including:

- Acute-to-Chronic Workload Ratio (ACR)
- Heart Rate / TSS
- Pace / Heart Rate
- Power / Heart Rate
- RPE / TSS

These features were designed to quantify how efficiently the athlete responded to training stress and to identify potential signs of overreaching or improved fitness.




7 Day Rolling TSS Over Time
7 Day Rolling TSS
0
50
150
250
350
0
20
40
60
80
100
Day


28 Day Rolling TSS Over Time
28 Day Rolling TSS
200
400
600
800
1000
0
20
40
60
80
100
Day

Exploratory Data Analysis

Exploratory analysis focused on identifying relationships between training load, physiological metrics, and HRV trends over time. Visual inspection of time-series data revealed clear fluctuations in HRV corresponding to periods of increased training load and recovery.

Correlation analysis indicated that many engineered features exhibited redundancy, particularly lagged TSS variables and rolling load metrics, which capture similar dynamics. As a result, feature selection emphasized simplicity and interpretability over maximum model complexity.

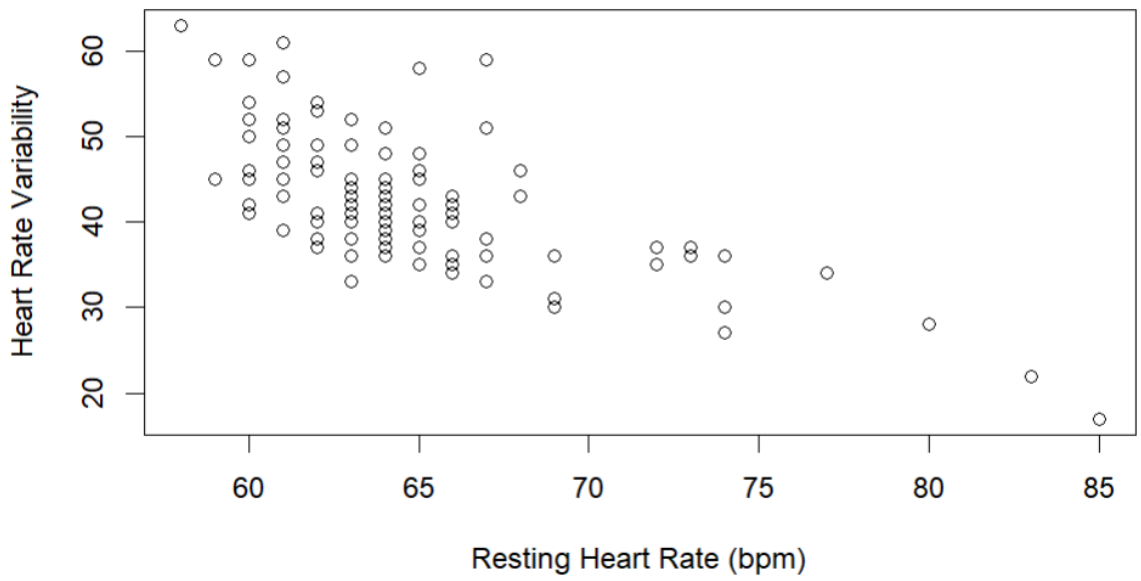
Regression Model

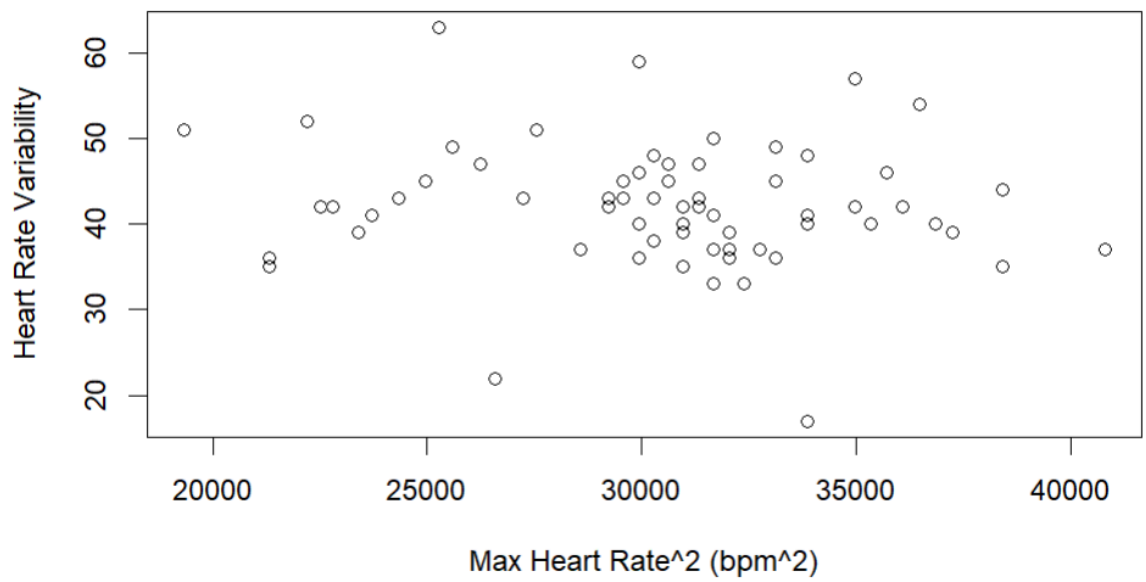
Following feature selection, a linear regression model was constructed to estimate HRV as a function of key physiological predictors:

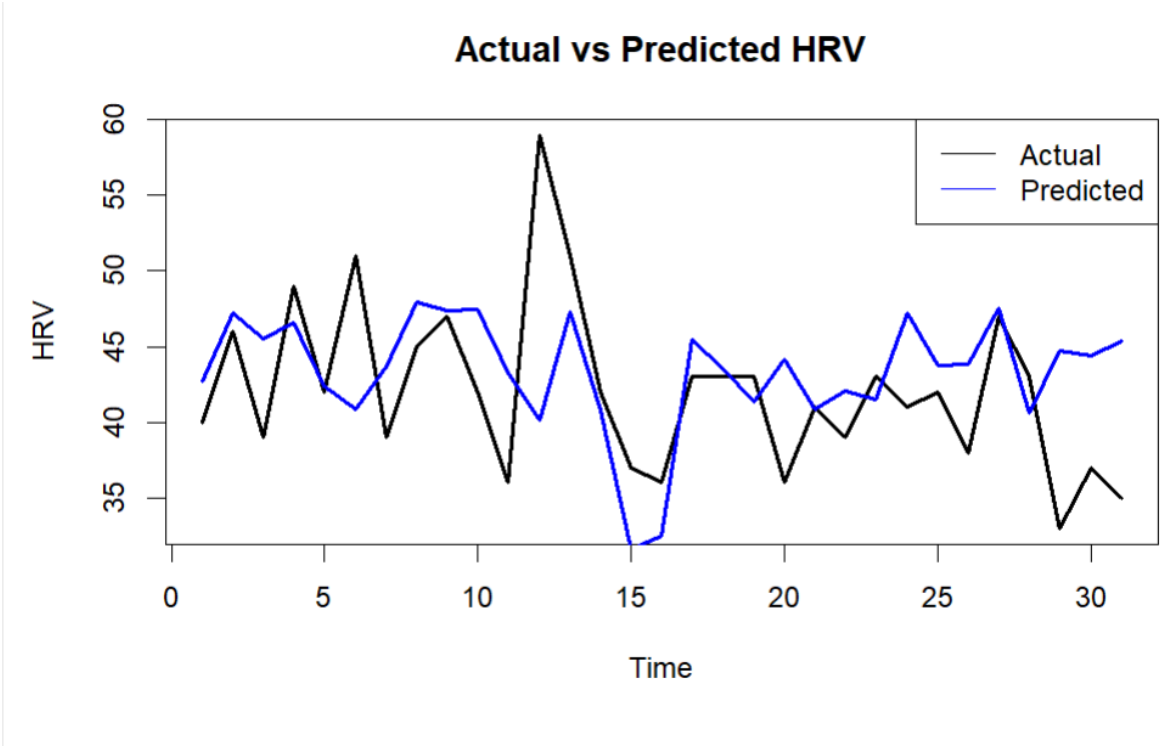
HRV~RHR+(Lagged Max HR)^2

This model was selected based on statistical significance, interpretability, and stability across subsets of the data. Both resting heart rate and lagged maximum heart rate demonstrated negative relationships with HRV, consistent with physiological expectations that elevated cardiovascular strain corresponds to a reduced recovery the following day.

Although the adjusted R^2 value was modest, this is typical in physiological modeling contexts, where outcome variables are influenced by numerous unobserved and stochastic factors. Individual t-tests indicated that both predictors contributed meaningfully to the model.


Heart Rate Variability
20
30
40
50
60
60
65
70
75
80
85
Resting Heart Rate (bpm)


Heart Rate Variability
20
30
40
50
60
20000
25000
30000
35000
40000
Max Heart Rate^2 (bpm^2)



Model Validation

To evaluate generalization performance, the dataset was split into training (65%) and testing (35%) subsets using a time-ordered split to preserve chronological structure.

Out-of-sample evaluation yielded:

- Test $R^2 \approx -0.2576$
- RMSE ≈ 6.0982

The negative out-of-sample R^2 indicates that the regression model does not outperform a naive mean-baseline on held-out-data, and this result deserves direct acknowledgment rather than dismissal. Several factors contribute to this outcome. First, with only 115 observations and a time-ordered split, the set covers a distinct temporal period with potentially different physiological dynamics than the training window – a distribution shift that linear models are particularly sensitive to. Second, HRV is an inherently high-noise signal influenced by variables

absent from the dataset entirely: hydration, psychological stress, illness, caffeine and sleep quality beyond raw duration. Third, the model's predictive utility was never intended to be absolute – it was designed to produce directional residuals for a decision system, not point forecasts. In that context, the RMSE of ~6 bpm against an HRV range of roughly 20-65 bpm represents a workable signal, even if standalone predictive accuracy is limited. Future iterations should consider nonlinear model architectures, a larger longitudinal dataset, and additional physiological covariates to address these constraints directly.

Decision System

To translate model outputs into actionable training guidance, a rule-based decision system was developed using residual analysis.

For each day:

1. HRV was predicted using the regression model
2. Residuals were computed as the difference between actual and predicted HRV
3. Residuals were standardized into z-scores

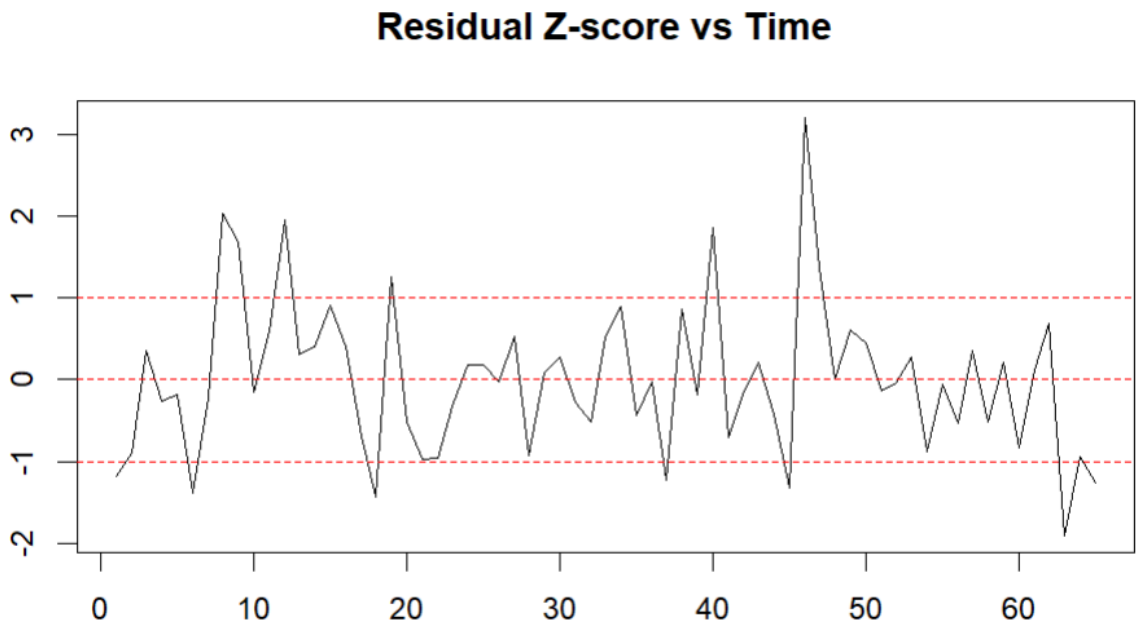
These z-scores represent deviations from expected recovery:

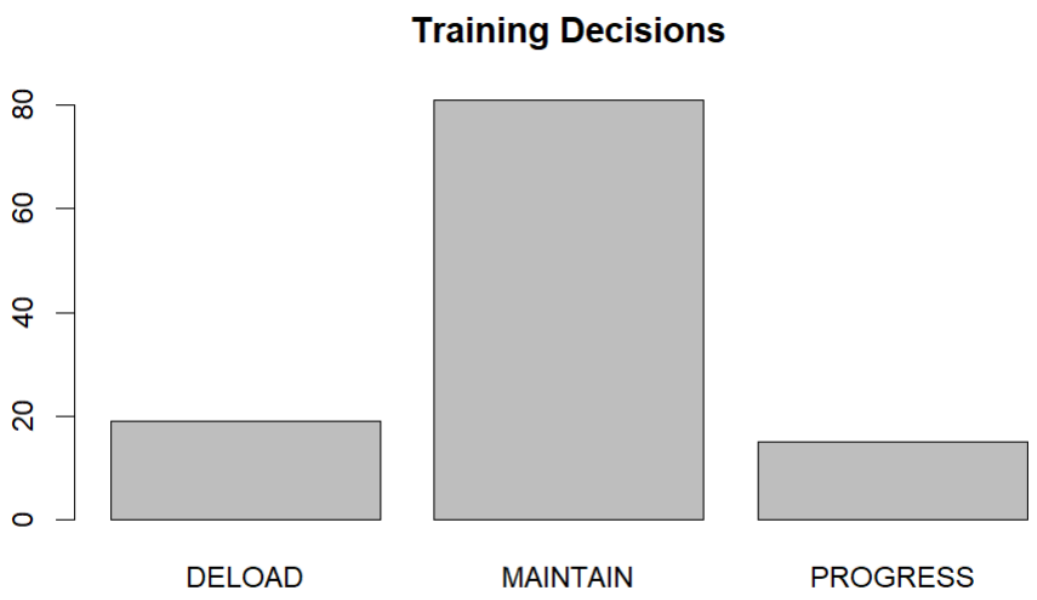
- Negative values indicate under-recovery
- Positive values indicate better-than-expected recovery

A three-zone decision framework was implemented:

- **DELOAD**: Residual z-score < -1
- **MAINTAIN**: $-1 \leq$ Residual z-score ≤ 1
- **PROGRESS**: Residual z-score > 1

This system enables dynamic adjustment of training intensity based on deviations from expected physiological response, rather than absolute values alone.


Residual Z-score vs Time
3
2
1
0
-1
-2
0
10
20
30
40
50
60


Training Decisions
80
60
40
20
0
DELOAD
MAINTAIN
PROGRESS

Results and System Behavior

Beginning October 25th, I began a structured triathlon-focused training block featuring approximately 15 sessions per week across all triathlon modalities and weightlifting. This protocol remained consistent across the entire study period through April 13th, providing a controlled baseline against which system performance could be evaluated.

The study was divided into two sequential blocks. Block 1 (October 25- February 13, approximately 16 weeks) consisted of structure training without prospective application of the decision system – this phase served as the data collection and model development period. Block 2 (February 13 - April 13, approximately 8 weeks) involved prospective, real-time application of the decision system to inform daily training adjustments under the same structured plan.

Performance outcomes across each block were as follows:

Block 1 (16 weeks, no decision system):

- Cycling FTP: +25W
- 5km run: -1:10
- Strength (5 standardized lifts): + 7.3%
- 100yd swim pace: -4s

Block 2 (8 weeks, decision system applied):

- Cycling FTP: +35W
- 5km run: -1:15
- Strength: +10.7%
- 100yd swim pace: -3s

The central finding is that Block 2 produced performance improvements that matched or exceeded those of Block 1 across nearly every metric, despite being approximately half the duration and covering the same structured training protocol. Critically, this occurred during a period when further improvement should be physiologically harder to achieve – adaptation gains

follow a logarithmic curve, meaning each additional increment of fitness requires disproportionately more stimulus. The fact that equivalent or greater gains were realized in half the time, against that biological headwind, provides meaningful evidence that the decision system was adding value beyond the training plan alone.

This is not a controlled experiment with a randomized comparison group, and that limitation should be acknowledged. Confounding factors – seasonal fitness momentum, improved technical efficiency, or psychological familiarity with the training protocol – cannot be fully ruled out. However, the consistency of the effect across four separate metrics, combined with the diminishing-returns context, lends credibility to the conclusion that residual-guided training adjustment contributed to the observed acceleration in adaptation.

DELOAD recommendations during Block 2 consistently followed periods of elevated acute load or suppressed recovery signals, while PROGRESS flags aligned with periods of strong HRV response limited to model expectations – a pattern consistent with the system functioning as intended.

Limitations

Several limitations should be noted:

- HRV is inherently noisy and influenced by factors not captured in the dataset (e.g., stress, illness, hydration, caffeine)
- Sample size was limited to 115 days
- Self-reported RPE introduces subjectivity
- Model structure was linear and may not capture nonlinear physiological relationships
- Missing or approximated values (e.g., max HR on non-training days) may introduce bias

Conclusion

This project demonstrates the development of a time-series analytics framework for modeling training load and recovery in an endurance athlete. While predictive accuracy of HRV

was limited, the transformation of model residuals into a structured decision system provided a practical method for interpreting physiological signals.

The integration of feature engineering, statistical modeling, and rule-based decision logic highlights the potential for data-driven approaches to inform training strategies in applied settings. Future work could incorporate larger datasets, nonlinear modeling techniques, and additional physiological variables to enhance predictive performance and system robustness.